Normal Equation

- The matrix (x^t)(x) becomes **singular (non-invertible)** or nearly singular.

  This means we cannot compute ((x^t)(x))^-1

- Because the columns of matrix XXX are **linearly dependent,** When features are linearly dependent:

➢ The matrix does not have full rank.

➢ Its determinant becomes zero.

➢ It cannot be inverted.

➢ There is no unique solution for the parameters.

- It can be solved using:

    - ➢ y adding λI to, making it invertible:

      ((x^t)(x)) so it will be

      ((x^t)(x)+ λI)^-1

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    - ➢ by using the **Moore–Penrose Pseudoinverse (SVD:Singular Value Decomposition)**

      The Moore–Penrose Pseudoinverse is a generalized matrix inverse computed using SVD. It allows us to compute regression parameters even when XTXX^T XXTX is singular. It provides the minimum-norm least squares solution without removing features.

Sources:

- https://en.wikipedia.org/wiki/Ordinary_least_squares
- https://en.wikipedia.org/wiki/Multicollinearity
- ChatGPT